

A character separating a closed ideal

MathlibAnnex.Analysis.CStarAlgebra.exists_character_annihilating_closedIdeal_of_not_mem

theorem

Finds one character which kills a closed ideal and detects a specified element outside it.

Statement

Let D be a commutative unital complex C^* -algebra, let J be a norm-closed ideal, and let $d \in D \setminus J$. There is a character $\chi : D \rightarrow \mathbb{C}$ such that

$$\chi(x) = 0 \quad (x \in J), \quad \chi(d) \neq 0.$$

A character here is a nonzero complex algebra homomorphism, hence unital and continuous.

Assumptions

Closedness is for the norm topology. Properness of J and nontriviality of D follow from $d \notin J$; they are not extra hypotheses.

Conclusion

The same character satisfies both annihilation of all of J and nonvanishing at d .

Proof route

Transport the ideal through the isometric Gelfand transform, find a point where the transformed element escapes its vanishing conditions, and evaluate there.

▼ Proof steps

1. Let X be the character space of D and $G : D \rightarrow C(X, \mathbb{C})$ its Gelfand $*$ -isomorphism, so $G(x)(\chi) = \chi(x)$. Put $K = G(J)$. Surjectivity and multiplicativity make K an ideal. Since G is an isometry and a homeomorphism, K is closed. If $G(d) \in K$, then $G(d) = G(x)$ for some $x \in J$, contradicting injectivity of G and $d \notin J$.
2. Apply The closed-ideal description applied to this Gelfand image to the norm-closed ideal K of continuous functions on the compact Hausdorff X . It gives

$$K = \{f \in C(X, \mathbb{C}) : f|_Z = 0\}, \quad Z = \{\chi \in X : g(\chi) = 0 \text{ for all } g \in K\}.$$

Since $G(d) \notin K$, there is $\chi \in Z$ with $G(d)(\chi) \neq 0$.

3. For $x \in J$, the function $G(x)$ belongs to K , hence $\chi(x) = G(x)(\chi) = 0$. At the distinguished element, $\chi(d) = G(d)(\chi) \neq 0$. These are the two required conditions on the very same point χ of the character space.

Main citations

- Exact theorem and proof

Lean source signature (exact)

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theorem exists_character_annihilating_closedIdeal_of_not_mem
  {D : Type u} [CommCStarAlgebra D]
  (J : Ideal D) (hJclosed : IsClosed (J : Set D))
  {d : D} (hd : d ∉ J) :
  ∃ chi : WeakDual.characterSpace C D,
    (∀ x : D, x ∈ J → chi x = 0) ∧ chi d ≠ 0
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In the source	Mathematical meaning
<code>{D : Type u} [CommCStarAlgebra D]</code>	The commutative unital complex C^* -algebra D .
<code>(J : Ideal D) (hJclosed : IsClosed (J : Set D))</code>	The ideal J and norm closedness of its set of elements.
<code>{d : D} (hd : d ∉ J)</code>	A specified element d outside that ideal.
<code>∃ chi : WeakDual.characterSpace C D</code>	Existence of one nonzero continuous complex character χ of D .
<code>(∀ x : D, x ∈ J → chi x = 0)</code>	This χ vanishes at every element belonging to J .
<code>∧ chi d ≠ 0</code>	The same χ is nonzero at the chosen d , in addition to all preceding vanishing equations.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.exists_character_annihilating_closedIdeal_of_not_mem

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